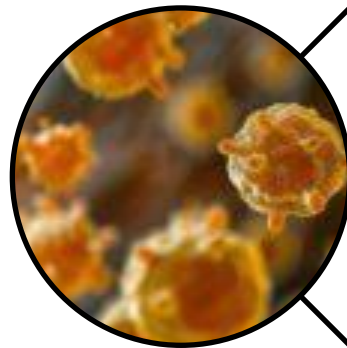


connecting people with science



**Antimicrobial
Resistance of Bacterial
Pathogens – Africa**

7-13 March 2026

NICD, Johannesburg,
South Africa

Collins Kigen

CP2 – Introduction to Commandline

March 2026

Navigation

Command	Description
	Print Current Working Directory
	List Files and Directories
	Long list + human-readable file size
	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
	List Files and Directories
	Long list + human-readable file size
	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
	List Files and Directories
	Long list + human-readable file size
	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
	Long list + human-readable file size
	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
	Long list + human-readable file size
	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
ls -a	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
ls -a	Show hidden files
	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
ls -a	Show hidden files
cd <dir>	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
ls -a	Show hidden files
cd <dir>	Change directory
	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
ls -a	Show hidden files
cd <dir>	Change directory
cd ..	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
ls -a	Show hidden files
cd <dir>	Change directory
cd ..	Move up one level
	Go to home directory

Navigation

Command	Description
pwd	Print Current Working Directory
ls	List Files and Directories
ls -lh	Long list + human-readable file size
ls -a	Show hidden files
cd <dir>	Change directory
cd ..	Move up one level
cd ~	Go to home directory

Files and Directory Management

Command	Description
	Create a directory
	Create empty file
	Open editor
	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
	Create empty file
	Open editor
	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
	Create empty file
	Open editor
	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
	Open editor
	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
	Open editor
	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
mv <src> <dest>	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
mv <src> <dest>	Move or rename file
	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
mv <src> <dest>	Move or rename file
rm <file>	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
mv <src> <dest>	Move or rename file
rm <file>	Remove File
	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
mv <src> <dest>	Move or rename file
rm <file>	Remove File
rm -r <dir>	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
mv <src> <dest>	Move or rename file
rm <file>	Remove File
rm -r <dir>	Remove Directory and File
	Copy File

Files and Directory Management

Command	Description
mkdir <dir>	Create a directory
touch <file>	Create empty file
nano <file>	Open editor
cp <src> <dest>	Copy File
mv <src> <dest>	Move or rename file
rm <file>	Remove File
rm -r <dir>	Remove Directory and File
cp <src> <dest>	Copy File

Viewing Files

Command	Description
	First 10 lines
	Last 10 line
	Scroll large files one page at a time
	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
<code>head -n 10 <file></code>	First 10 lines
	Last 10 line
	Scroll large files one page at a time
	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
head -n 10 <file>	First 10 lines
	Last 10 line
	Scroll large files one page at a time
	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
head -n 10 <file>	First 10 lines
tail -n 10 <file>	Last 10 line
	Scroll large files one page at a time
	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
<code>head -n 10 <file></code>	First 10 lines
<code>tail -n 10 <file></code>	Last 10 line
	Scroll large files one page at a time
	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
<code>head -n 10 <file></code>	First 10 lines
<code>tail -n 10 <file></code>	Last 10 line
<code>less <file></code>	Scroll large files one page at a time
	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
<code>head -n 10 <file></code>	First 10 lines
<code>tail -n 10 <file></code>	Last 10 line
<code>less <file></code>	Scroll large files one page at a time
	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
<code>head -n 10 <file></code>	First 10 lines
<code>tail -n 10 <file></code>	Last 10 line
<code>less <file></code>	Scroll large files one page at a time
<code>more <file></code>	Similar to less but only forward move
	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
<code>head -n 10 <file></code>	First 10 lines
<code>tail -n 10 <file></code>	Last 10 line
<code>less <file></code>	Scroll large files one page at a time
<code>more <file></code>	Similar to less but only forward move
	View entire file at once
	View compressed file like <code>sample.fastq.gz</code>

Viewing Files

Command	Description
head -n 10 <file>	First 10 lines
tail -n 10 <file>	Last 10 line
less <file>	Scroll large files one page at a time
more <file>	Similar to less but only forward move
cat <file>	View entire file at once
	View compressed file like sample.fastq.gz

Viewing Files

Command	Description
<code>head -n 10 <file></code>	First 10 lines
<code>tail -n 10 <file></code>	Last 10 line
<code>less <file></code>	Scroll large files one page at a time
<code>more <file></code>	Similar to less but only forward move
<code>cat <file></code>	View entire file at once
	View compressed file like <code>sample.fastq.gz</code>

Viewing Files

Command	Description
head -n 10 <file>	First 10 lines
tail -n 10 <file>	Last 10 line
less <file>	Scroll large files one page at a time
more <file>	Similar to less but only forward move
cat <file>	View entire file at once
zcat <file.gz>	View compressed file like sample.fastq.gz

Pipe, Redirect, Append

Command	Description
	sends standard output (stdout) to another command's standard input
	Redirection sends the output of a command to a file
	Redirects stdout to a file, appending the output to the end of the file

Pipe, Redirect, Append

Command	Description
command1 command2	sends standard output (stdout) to another command's standard input
	Redirection sends the output of a command to a file
	Redirects stdout to a file, appending the output to the end of the file

Pipe, Redirect, Append

Command	Description
command1 command2	sends standard output (stdout) to another command's standard input
	Redirection sends the output of a command to a file
	Redirects stdout to a file, appending the output to the end of the file

Pipe, Redirect, Append

Command	Description
command1 command2	sends standard output (stdout) to another command's standard input
echo "This is new content" > file.txt	Redirection sends the output of a command to a file
	Redirects stdout to a file, appending the output to the end of the file

Pipe, Redirect, Append

Command	Description
command1 command2	sends standard output (stdout) to another command's standard input
echo "This is new content" > file.txt	Redirection sends the output of a command to a file
echo "This content will be	Redirects stdout to a file, appending the output to the end of the file

Pipe, Redirect, Append

Command	Description
command1 command2	sends standard output (stdout) to another command's standard input
echo "This is new content" > file.txt	Redirection sends the output of a command to a file
echo "This content will be appended" >> file.txt	Redirects stdout to a file, appending the output to the end of the file

Searching and counting

Command	Description
	Count lines
	Count characters
	Search for text
	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
wc -l	Count lines
	Count characters
	Search for text
	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
wc -l	Count lines
	Count characters
	Search for text
	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
wc -l	Count lines
wc -c	Count characters
	Search for text
	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
wc -l	Count lines
WC -c	Count characters
	Search for text
	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
wc -l	Count lines
wc -c	Count characters
grep "pattern" <file>	Search for text
	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
wc -l	Count lines
wc -c	Count characters
grep "pattern" <file>	Search for text
	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
<code>wc -l</code>	Count lines
<code>wc -c</code>	Count characters
<code>grep "pattern" <file></code>	Search for text
<code>grep -c "pattern" <file></code>	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
wc -l	Count lines
wc -c	Count characters
grep "pattern" <file>	Search for text
grep -c "pattern" <file>	Count the number of occurrences
	Sort and count unique

Searching and counting

Command	Description
<code>wc -l</code>	Count lines
<code>wc -c</code>	Count characters
<code>grep "pattern" <file></code>	Search for text
<code>grep -c "pattern" <file></code>	Count the number of occurrences
<code>sort file.txt uniq -c</code>	Sort and count unique

Productivity Shortcuts

Command	Description
	Autocomplete
	Stop running process
	Clear screen
	Previous command
	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
	Stop running process
	Clear screen
	Previous command
	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
	Stop running process
	Clear screen
	Previous command
	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
	Clear screen
	Previous command
	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
	Clear screen
	Previous command
	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
	Previous command
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	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
	Previous command
	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
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Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
	Previous command
	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
↑	Previous command
Ctrl + R	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
↑	Previous command
Ctrl + R	Search command history
	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
↑	Previous command
Ctrl + R	Search command history
!!	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
↑	Previous command
Ctrl + R	Search command history
!!	Run last command
	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
↑	Previous command
Ctrl + R	Search command history
!!	Run last command
Ctrl + a	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
↑	Previous command
Ctrl + R	Search command history
!!	Run last command
Ctrl + a	Go to the start of the command line
	Go to the end of the command line

Productivity Shortcuts

Command	Description
Tab	Autocomplete
Ctrl + C	Stop running process
Ctrl + L	Clear screen
↑	Previous command
Ctrl + R	Search command history
!!	Run last command
Ctrl + a	Go to the start of the command line
Ctrl + e	Go to the end of the command line

Help

Command	Description
man <command>	Full manual for any command
<command> --help	Quick Help
history	Show previous commands that have been run

Acknowledgements and Partners



thank you!

Please contact wellcomeconnectingscience.org
for more information.

